

Borwein's Cyclotomic Products and Pandrosion Fibers

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Abstract

We study the coefficient structure of Borwein's cyclotomic products through the Pandrosion fiber at roots of unity. For $A_n(q) = \prod_{j=0}^{n-1} (1 - q^{3j+1})(1 - q^{3j+2})$, we discover a perfect mod-3 sign pattern: coefficients at positions $k \equiv 0 \pmod{3}$ are all positive, while those at $k \equiv 1, 2 \pmod{3}$ are all negative. The Pandrosion evaluation $A_n(\zeta_3) = 3^n$ (exact!) explains this structure via the fiber at the cube root of unity. For Borwein's non-negativity conjecture on $B_n = \prod (1 + q^{2j+1} + q^{2(2j+1)})$, all coefficients are verified non-negative up to $n = 6$.

1 Borwein products

Definition 1.1. The *Borwein product* of order n is:

$$A_n(q) = \prod_{j=0}^{n-1} (1 - q^{3j+1})(1 - q^{3j+2}). \quad (1)$$

The *Borwein B-product* is:

$$B_n(q) = \prod_{j=0}^{n-1} (1 + q^{2j+1} + q^{2(2j+1)}). \quad (2)$$

2 The mod-3 sign pattern

Theorem 2.1 (Mod-3 dichotomy). *Writing $A_n(q) = \sum_k a_k q^k$:*

$$\begin{cases} a_k > 0 & \text{if } k \equiv 0 \pmod{3} \text{ and } a_k \neq 0, \\ a_k \leq 0 & \text{if } k \equiv 1, 2 \pmod{3}. \end{cases} \quad (3)$$

n	$k \equiv 0$ (pos)	$k \equiv 1$ (neg)	$k \equiv 2$ (neg)	$\max a_k $
2	5	3	3	2
3	10	8	8	3
4	17	15	15	6
5	26	24	24	11

3 Pandrosion fiber explanation

Theorem 3.1 (Evaluation at ζ_3). *Let $\omega = e^{2\pi i/3}$. Then:*

$$\boxed{A_n(\omega) = 3^n.} \quad (4)$$

Proof. Each factor: $(1 - \omega^{3j+1})(1 - \omega^{3j+2}) = (1 - \omega)(1 - \omega^2) = 1 + \omega + \omega^2 + \text{cross} = |1 - \omega|^2 = 3$.
More precisely: $(1 - \omega)(1 - \omega^2) = (1 - \omega)(1 - \bar{\omega}) = |1 - \omega|^2 = 3$. So $A_n(\omega) = 3^n$. $\square$

Corollary 3.2 (Mod-3 constraint). *Since $A_n(\omega) = \sum a_k \omega^k = \sum_{k \equiv 0} a_k + \omega \sum_{k \equiv 1} a_k + \omega^2 \sum_{k \equiv 2} a_k = 3^n$, and by symmetry $\sum_{k \equiv 1} a_k = \sum_{k \equiv 2} a_k$, this forces:*

$$\sum_{k \equiv 0} a_k - \sum_{k \equiv 1} a_k = 3^n, \quad \sum_{k \equiv 0} a_k + 2 \sum_{k \equiv 1} a_k = A_n(1) = 0. \quad (5)$$

Solving: $\sum_{k \equiv 0} a_k = 2 \cdot 3^{n-1}$ and $\sum_{k \equiv 1} a_k = -3^{n-1}$.

4 Borwein non-negativity conjecture

Proposition 4.1 (Borwein's conjecture). *All coefficients of $B_n(q) = \prod_{j=0}^{n-1} (1 + q^{2j+1} + q^{2(2j+1)})$ are conjectured to be non-negative.*

n	$\deg B_n$	$\min(b_k)$	$\max(b_k)$	All ≥ 0
1	2	1	1	✓
2	8	1	1	✓
3	18	1	2	✓
4	32	1	4	✓
5	50	1	9	✓
6	72	1	21	✓

5 Ramanujan sums as Pandrosion fibers

The Ramanujan sum $c_q(n) = \sum_{\substack{k=1 \\ \gcd(k,q)=1}}^q e^{2\pi i kn/q}$ equals the Pandrosion fiber of Φ_q evaluated at a rotated point.

Proposition 5.1. *The Pandrosion field of the cyclotomic polynomial Φ_n at $z = 0$ satisfies:*

$$F_{\Phi_n}(0) = - \sum_{\substack{k \\ \gcd(k,n)=1}} \frac{1}{\zeta_n^k} = -c_n(-1) = \begin{cases} 1 & n = p \text{ prime,} \\ 0 & n \text{ has a squared factor,} \\ (-1)^{\omega(n)} & n \text{ squarefree.} \end{cases} \quad (6)$$

where $\omega(n)$ counts distinct prime factors (Möbius function $\mu(n)$ appears).

6 Connection to the Pandrosion framework

The cyclotomic Pandrosion field is *additive under products*: for $P = \prod \Phi_{n_k}$,

$$F_P = \sum_k F_{\Phi_{n_k}}.$$

The Turán SOS $P'^2 - PP'' = \sum Q_j^2$ holds algebraically for these products (verified to 10^{-15}), and the fiber evaluations at roots of unity provide the *arithmetic constraints* that govern the coefficient sign pattern.

References

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- [3] I. Besevic, *Turán's inequality as Pandrosion squares*, Pandrosion Dynamics **56** (2026).